



3- Course Handbook - Topics in Software Engineering I

Software Engineering (COMSATS University Islamabad)



Scan to open on Studocu



COMSATS University Islamabad, Lahore Campus
DEPARTMENT OF COMPUTER SCIENCE

COURSE HANDBOOK

1	Course Title	Topics in Software Engineering I (Software Re-Engineering)	
2	Course Code	CSE 350	
3	Credit Hours	3(3,0)	
4	Semester	Fall 2022	
5	Resource Person	Dr. Fatima Sabir	
6	Supporting Team Members	Nil	
7	Contact Hours (Theory)	3 hours per week	
8	Contact Hours (Lab)	Nil	
9	Office Hours	Thur:10:30 to 11:30am	
10	Course Introduction		
This course explains concepts and technique of software re-engineering; apply reengineering techniques to maintain and modify software systems; to analyze and understand maintenance related problems associated with object oriented software systems; to perform complex design reengineering and reverse engineering problems.			
11	Learning Objectives		
Subject Specific: Knowledge, Understanding and Skills			
- To explain the concepts and technique of software re-engineering; - To apply reengineering techniques to maintain and modify software systems; - To analyze and understand maintenance related problems associated with object oriented software systems; - To perform complex design reengineering and reverse engineering problems.			
12	Course Contents		
This course will focus on salient topics include the terminology and the processes pertaining to software evolution, fundamental re- engineering techniques to modernize legacy systems including source code analysis, architecture recovery, and code restructuring, software refactoring strategies, migration to Object Oriented platforms, quality issues in re- engineering processes, migration to network-centric environments, and software integration, reverse engineering, program comprehension, source code transformation and refactoring strategies, software maintenance and re-engineering economics.			
13	Lecture Schedule		
Weeks	Topic of Lecture		
Week 1	Software Re-Engineering: Introduction, Evolution, and Legacy Systems.		
Week 2	Techniques of Re-Engineering: Reverse Engineering, Restructuring, and Forward Propagation.		

Week 3	Techniques of Re-Engineering: Reverse Engineering, Restructuring, and Forward Propagation.	
Week 4	Techniques of Re-Engineering: Reverse Engineering, Restructuring, and Forward Propagation. Refactoring & Applications; Re-Architecting; and Big Rewrite.	
Week 5	Refactoring & Applications; Re-Architecting; and Big Rewrite.	
Week 6	Refactoring & Applications; Re-Architecting; and Big Rewrite.	
Week 7	Project Improvement & Automating the Development Environment	
Week 8	Project Improvement & Automating the Development Environment	
Week 9	Object Oriented Re-engineering Patterns	
Week 10	Object Oriented Re-engineering Patterns, Quality Issues in Re-Engineering Processes	
Week 11	Quality Issues in Re-Engineering Processes	
Week 12	Tool Support for Re-engineering; and Challenges & Stakeholder Aspiration	
Week 13	Tool Support for Re-engineering; and Challenges & Stakeholder Aspiration	
Week 14	Software Maintenance; and Re-engineering Economics	
Week 15	Software Maintenance; and Re-engineering Economics	
Week 16	Software Maintenance; and Re-engineering Economics	

14 Course Assessment

The assessment of this module shall have following breakdown structure

Mid term Exam	25%
Quizzes	15%
Assignments	10%
Terminal Examination	50%

The minimum pass marks for each course shall be 50%. Students obtaining less than 50% marks in any course shall be deemed to have failed in that course. The correspondence between letter grades, credit points, and percentage marks at CIIT shall be as follows:

Grades	Letter Grade	Credit Points	Percentage Marks
A	(Excellent)	4.0	90and above
A-		3.7	85-89
B+		3.3	80-84
B	(Good)	3.0	75-79
B-		2.7	70-74
C+		2.3	65-69
C	(Average)	2.0	60-64
C-		1.7	55-59
D	(Minimum passing)	1.3	50-54
F	(Failing)	0.0	Less than 50

Note: The marks to be assigned to students shall be in whole numbers and are not same as followed in the annual system of Lancaster University.

15 Assessment Schedule

Week 2 | 1st Assignment

	Week 3	1 st Quiz
	Week 9	2 nd Assignment
	Week 10	2 nd Quiz
	Week 13	3 rd Assignment
	Week 14	3 th Quiz
	Week 16	Final Presentation

16.	Format of Assignment
------------	-----------------------------

17.	Text Book	1. Re-engineering legacy software, David Lorge Parnas, Chris Birchall, Safari Books, Shelter Island, NY, 2016 2. Reengineering, Priyadarshi Tripathy and Kshirasagar Naik, John Wiley & Sons, Inc.2015
------------	------------------	---

18.	Reference Books	Software Maintenance and Evolution: a Roadmap, K.H.Bennett and V.T Rajlich, The Future of Software Engineering, ACM Press 2000.
------------	------------------------	---

19.	Plagiarism
------------	-------------------

Plagiarism involves the unacknowledged use of someone else's work, usually in coursework, and passing it off as if it were one's own. Many students who submit apparently plagiarised work probably do so inadvertently without realising it because of poorly developed study skills, including note taking, referencing and citations; this is poor academic practice rather than malpractice. Some students, particularly those from different cultures and educational systems, find UK academic referencing/acknowledgement systems and conventions awkward, and proof-reading is not always easy for dyslexic students and some visually-impaired students. Study skills education within programmes of study should minimise the number of students submitting poorly referenced work. However, some students plagiarise deliberately, with the intent to deceive. This intentional malpractice is a conscious, pre-mediated form of cheating and is regarded as a particularly serious breach of the core values of academic integrity.

Plagiarism can include the following:

1. collusion, where a piece of work prepared by a group is represented as if it were the student's own;
2. commission or use of work by the student which is not his/her own and representing it as if it were, e.g.:
 - a. purchase of a paper from a commercial service, including internet sites, whether pre-written or specially prepared for the student concerned
 - b. submission of a paper written by another person, either by a fellow student or a person who is not a member of the university;
3. duplication (of one's own work) of the same or almost identical work for more than one module;
4. the act of copying or paraphrasing a paper from a source text, whether in manuscript, printed or electronic form, without appropriate acknowledgement (this includes quoting directly from another source with a reference but without quotation marks);
5. submission of another student's work, whether with or without that student's knowledge or consent;
6. Directly quoting from model solutions/answers made available in previous years;
7. cheating in class tests, e.g.
 - a. when a candidate communicates, or attempts to communicate, with a fellow candidate or individual who is neither an invigilator or member of staff
 - b. copies, or attempts to copy from a fellow candidate

- c. attempts to introduce or consult during the examination any unauthorised printed or written material, or electronic calculating, information storage device, mobile phones or other communication device
 - d. personates or allows himself or herself to be impersonated.
8. Fabrication of results occurs when a student claims to have carried out tests, experiments or observations that have not taken place or presents results not supported by the evidence with the object of obtaining an unfair advantage.

These definitions apply to work in whatever format it is presented, including written work, online submissions, groupwork and oral presentations.

20.	Attendance Policy
------------	--------------------------

	Every student must attend 80% of the lectures/seminars delivered in each course and 80% of the practical/laboratory work prescribed for the respective courses. The students falling short of required percentage of attendance of lectures/seminars/practical/laboratory work, etc., shall not be allowed to appear in the terminal examination of this course and shall be treated as having failed this course.
--	--

21.	Field Trips/Case Studies/Seminars/Workshop
------------	---